

Mathematics

20/8/23

$$1) \triangle ABC \sim \triangle DCB$$

$$\frac{AB}{DC} = \frac{AC}{DB} \Rightarrow AB \times DB = AC \times DC$$

$$\therefore \text{Ans: } AC \times DC$$

$$2) A(a, b), O(0, 0)$$

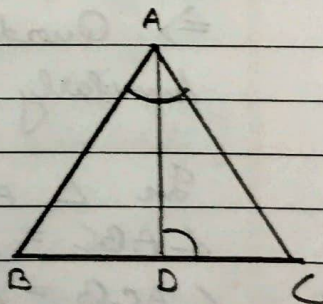
$$AO = \sqrt{(a-0)^2 + (b-0)^2} \\ = \sqrt{a^2 + b^2}$$

$$\therefore \text{Ans: } \sqrt{a^2 + b^2}$$

$$3) \text{ In } \triangle ABC \text{ and } \triangle ADC \\ \angle BAC = \angle ADC \text{ [given]} \\ \angle ACB = \angle ACD \text{ [common]}$$

$$\therefore \triangle ABC \sim \triangle DAC \text{ [AA]} \\ \Rightarrow \frac{CA}{CB} = \frac{AB}{AD} = \frac{AC}{CD}$$

$$\therefore \text{Ans: } AB \times CD = AC \times AD$$



$$4) \text{ Let the points be } A(x_1, y_1) \text{ and } B(x_2, y_2) \\ P(x, y) \text{ divides } AB \text{ in the ratio } m_1 : m_2$$

$$x \text{ of } P = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

$$y \text{ of } P = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

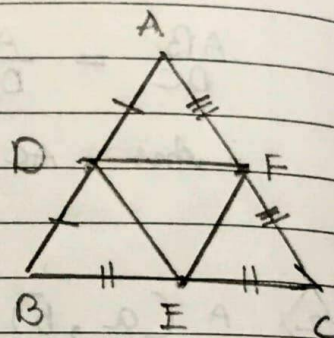
5) In $\triangle ABC$

$AD = BD$ [D is the midpoint]

$AF = CF$ [F is the midpoint]

$\therefore DF \parallel BC$ and $BC = 2DF$

[By Midpoint theorem]



$$DF = \frac{1}{2} BC$$

$\Rightarrow DF = BE$ [E is the midpoint of BC]

Since $DF = BE$ and $DF \parallel BE$

$\Rightarrow$ Quadrilateral BDEF is a $\parallel^m$

Similarly DFCE is a $\parallel^m$

In $\triangle ABC$ and $\triangle EDF$

$\angle ABC = \angle EFD$ [opposite $\angle$ s in a $\parallel^m$]

$\angle ACB = \angle FED$ [opposite $\angle$ s in a $\parallel^m$]

$\therefore \triangle ABC \sim \triangle EFD$ [AA similarity]

If two triangles are similar, then the ratio of their areas is equal to the ratio of the squares of their corresponding sides

$$\therefore \frac{\text{ar}(\triangle EDF)}{\text{ar}(\triangle ABC)} = \left(\frac{DF}{BC}\right)^2 = \left(\frac{DF}{2DF}\right)^2 = \frac{1}{4}$$

$\therefore$ The ratio of their areas is 1:4

- 6) Let the points be $A(-3, 2)$ and $B(6, 1)$
 Let the point which divides ~~the~~ AB by the y -axis
 be $P(0, y)$
 Let the ratio be $m_1 : m_2$

$$x \text{ of } P = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

$$0 = \frac{6m_1 + (-3)m_2}{m_1 + m_2}$$

$$2m_1 = 3m_2$$

$$\frac{m_1}{m_2} = \frac{3}{2}$$

$\therefore$ The point P divides AB in the ratio $3:2$.

- 7) In $\triangle ABC$ and $\triangle ADE$
 $\angle BAC = \angle DAE$ [common]
 $DE \parallel BC$, AB transversal
 $\angle ABC = \angle ADE$ [corresponding $\angle$ s]

$\therefore \triangle ABC \sim \triangle ADE$ [AA similarity]

$$\Rightarrow \frac{AD}{AB} = \frac{AE}{AC}$$

$$\Rightarrow \frac{x+4}{2x+7} = \frac{2x-1}{3x} \Rightarrow 3x^2 + 12x = 4x^2 - 2x + 14x - 7$$

$$\Rightarrow 10x = x^2 + 14x - 7$$

$$\Rightarrow x^2 + 4x - 7 = 0$$

$$\Rightarrow x^2 = 7$$

$$\Rightarrow x = \sqrt{7}$$

$\therefore$ The value of x is $\sqrt{7}$

- 8) Let the points be $A(-3, 4)$, $B(2, 5)$ and $P(x, 0)$

$$AP = BP$$

$$AP^2 = BP^2$$

$$(-3-x)^2 + (4-0)^2 = (2-x)^2 + (5-0)^2$$

$$9 + x^2 + 6x + 16 = 4 + x^2 - 4x + 25$$

$$10x = 4$$

$$x = 2/5$$

$\therefore$ The point is $(2/5, 0)$

- 9) In $\triangle ABD$

$$AD^2 = AB^2 + BD^2$$

$$AD^2 = AB^2 + \left(\frac{BC}{2}\right)^2 \quad [\because BD = DC]$$

$$AD^2 = AB^2 + \frac{BC^2}{4} \quad \text{--- (i)}$$

In $\triangle BCE$

$$CE^2 = BC^2 + BE^2$$

$$CE^2 = BC^2 + \frac{AB^2}{4} \quad [\because BE = EA] \quad \text{--- (ii)}$$

Add (i) and (ii)

$$AD^2 + CE^2 = AB^2 + \frac{1}{4}BC^2 + BC^2 + \frac{1}{4}AB^2$$

$$AD^2 + CE^2 = \frac{5}{4}(AB^2 + BC^2)$$

$$AD^2 + CE^2 = \frac{5}{4}AC^2 \quad [\because AC^2 = AB^2 + BC^2]$$

$$\left(\frac{3\sqrt{5}}{2}\right)^2 + CE^2 = \frac{5}{4} \times 25$$

$$CE^2 = \frac{125}{4} - \frac{45}{4} = 20$$

$$CE = \sqrt{20} = 2\sqrt{5} \text{ cm}$$

$\therefore$ The length of CE is $2\sqrt{5} \text{ cm}$

10) $A(6, 8), B(3, 7), C(-2, -2), D(1, -1)$

$$AB = \sqrt{(6-3)^2 + (8-7)^2} = \sqrt{10}$$

$$BC = \sqrt{(3+2)^2 + (7+2)^2} = \sqrt{106}$$

$$CD = \sqrt{(-2-1)^2 + (-2+1)^2} = \sqrt{10}$$

$$AD = \sqrt{(6-1)^2 + (8+1)^2} = \sqrt{106}$$

$$AC = \sqrt{(6+2)^2 + (8+2)^2} = 2\sqrt{41}$$

$$BD = \sqrt{(3-1)^2 + (7+1)^2} = \sqrt{68}$$

$$AB = CD \text{ and } BC = AD \text{ and } AC \neq BD$$

$\therefore$ ABCD is a //gm